

DAY-4

TASK 1: IMPROVED DEMO VERSION OF POSE DETECTION USING LANDMARKS:

Today, I improved the existing pose detection by optimizing MediaPipe confidence settings and using a higher complexity pose model for better accuracy.

IMPROVED POSE DETECTION CODE:

```

improvepose.py > ...
1  import cv2
2  import mediapipe as mp
3  import time
4  import math
5  mp_pose = mp.solutions.pose
6  mp_drawing = mp.solutions.drawing_utils
7  pose = mp_pose.Pose(
8      static_image_mode=False,
9      model_complexity=2,
10     smooth_landmarks=True,
11     min_detection_confidence=0.7,
12     min_tracking_confidence=0.7
13 )
14 cap = cv2.VideoCapture(0)
15 cap.set(3, 1280)
16 cap.set(4, 720)
17 previous_time = 0
18 while True:
19     success, frame = cap.read()
20     if not success:
21         break
22     frame = cv2.flip(frame, 1)
23     frame = cv2.convertScaleAbs(frame, alpha=1.2, beta=20)
24     rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
25     results = pose.process(rgb_frame)
26     if results.pose_landmarks:
27         mp_drawing.draw_landmarks(
28             frame,
29             results.pose_landmarks,
30             mp_pose.POSE_CONNECTIONS,
31             mp_drawing.DrawingSpec(
32                 color=(0, 255, 0),
33                 thickness=2,
34                 circle_radius=2
35             ),
36             mp_drawing.DrawingSpec(
37                 color=(255, 0, 0),
38                 thickness=2
39             )
40         )
41     landmarks = results.pose_landmarks.landmark
42     left_shoulder = landmarks[mp_pose.PoseLandmark.LEFT_SHOULDER.value]

```

```

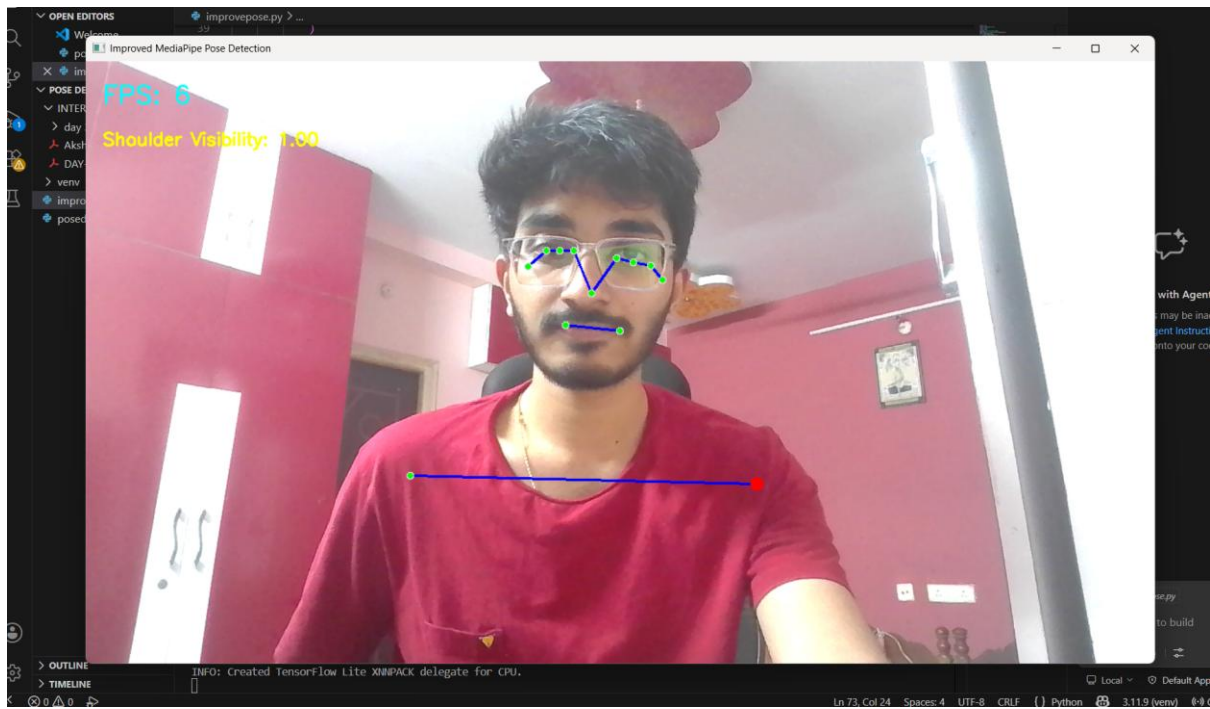
43     visibility = left_shoulder.visibility
44     cv2.putText(
45         frame,
46         f'Shoulder Visibility: {visibility:.2f}',
47         (20, 100),
48         cv2.FONT_HERSHEY_SIMPLEX,
49         0.7,
50         (0, 255, 255),
51         2
52     )
53     h, w, c = frame.shape
54     cx = int(left_shoulder.x * w)
55     cy = int(left_shoulder.y * h)
56     cv2.circle(frame, (cx, cy), 8, (0, 0, 255), cv2.FILLED)
57     current_time = time.time()
58     fps = 1 / (current_time - previous_time)
59     previous_time = current_time
60     cv2.putText(
61         frame,
62         f'FPS: {int(fps)}',
63         (20, 50),
64         cv2.FONT_HERSHEY_SIMPLEX,
65         1,
66         (255, 255, 0),
67         2
68     )
69     cv2.imshow("Improved MediaPipe Pose Detection", frame)
70     if cv2.waitKey(1) & 0xFF == ord('q'):
71         break
72 cap.release()
73 cv2.destroyAllWindows()

```

Additional enhancements such as landmark smoothing, HD camera resolution, and lighting adjustments were implemented to achieve more stable and precise body tracking. Visibility detection and FPS monitoring were also added to improve performance analysis and real-time tracking efficiency.

These improvements make the system more suitable for AI-based virtual try-on and body measurement applications.

OUTPUT:



CHANGES IN CODE:

1) Improved Confidence Settings:

Reduces false body detections

CODE:

```
min_detection_confidence=0.7,
```

```
min_tracking_confidence=0.7
```

2. Heavy Pose Model

Better landmark precision

CODE:

```
model_complexity=2
```

3. Landmark Smoothing

Reduces shaking landmarks

CODE:

```
smooth_landmarks=True
```

4. Increased Camera Resolution

Better image quality

CODE:

```
cap.set(3, 1280)
```

```
cap.set(4, 720)
```

5. Mirror View

Easier user interaction

CODE:

```
frame = cv2.flip(frame, 1)
```

6. Brightness Enhancement

Better detection in low lighting

CODE:

```
frame = cv2.convertScaleAbs(frame, alpha=1.2, beta=20)
```

7. Visibility Score Detection

Each landmark gets a visibility score:

CODE:

```
visibility = left_shoulder.visibility
```

8. FPS Counter

Real-time processing speed

CODE:

```
fps = 1 / (current_time - previous_time)
```